


**Figure 10-3** Satellite of mass $m_s$ in orbit around Earth. For the orbit to be geostationary, the distance $R_0$ between the satellite and Earth's center should be 42,164 km. At the equator, this corresponds to an altitude of 35,786 km above Earth's surface.

$$F_c = \frac{M_s u_s^2}{R_0} = M_s \omega^2 R_0, \tag{10.2}$$

where $G = 6.67 \times 10^{-11}$ N·m$^2$/kg$^2$ is the universal gravitational constant, $M_e = 5.98 \times 10^{24}$ kg is Earth's mass, $R_0$ is the distance between the satellite and the center of Earth, and $u_s$ is the satellite velocity. For a rotating object, $u_s = \omega R_0$, where $\omega$ is its angular velocity. In order for the satellite to remain in orbit, the two opposing forces acting on it have to be equal in magnitude, or

$$G\frac{M_s M_e}{R_0^2} = M_s \omega^2 R_0, \tag{10.3}$$

which yields a solution for $R_0$ given by

$$R_0 = \left(\frac{GM_e}{\omega^2}\right)^{1/3}. \tag{10.4}$$

To remain stationary with respect to Earth's surface, the satellite's angular velocity has to be the same as that of Earth's own angular velocity around its own axis. Thus,

$$\omega = \frac{2\pi}{T}, \tag{10.5}$$

where $T$ is the period of one sidereal day in seconds. A sidereal day, which takes into account Earth's rotation around the sun, is equal to 23 hours, 56 minutes, and 4.1 seconds. Using Eq. (10.5) in Eq. (10.4) gives

$$R_0 = \left(\frac{GM_e T^2}{4\pi^2}\right)^{1/3}, \tag{10.6}$$

and upon using the numerical values for $T$, $M_e$, and G, we obtain the result $R_0 = 42{,}164$ km. Subtracting 6,378 km for Earth's mean radius at the equator gives an altitude of $h = 35{,}786$ km above Earth's surface.

From a geostationary orbit, Earth subtends an angle of 17.4°, covering an arc of about 18,000 km along the equator, which corresponds to a longitude angle of about 160°. With three equally spaced satellites in geostationary orbit over the Earth's equator, it is possible to achieve complete global coverage of the entire equatorial plane with significant overlap between the beams of the three satellites. Concerning coverage toward the poles, a global beam can reach Earth stations up to 81° of latitude on either side of the equator.

Not all satellite communication systems use spacecraft that are in geostationary orbits. Indeed, because of transmitter power limitations or other considerations, it is sometimes necessary to operate from much lower altitudes. In this case, the satellite is placed in a ***highly elliptical orbit*** (to satisfy Kepler's law) so that for part of the orbit (near its perigee) it is at a range of only a few hundred kilometers from Earth's surface. Whereas only three geostationary satellites are needed to provide near-global coverage of Earth's surface, a much larger number is needed when the satellites operate from highly elliptical orbits. A good example of the latter is the Global Positioning System (GPS) described in Technology Brief 5.

## 10-2 Satellite Transponders

A communication satellite functions as a distant repeater; it receives ***uplink*** signals from Earth stations, processes the signals, and then ***downlinks*** (retransmits) them to their



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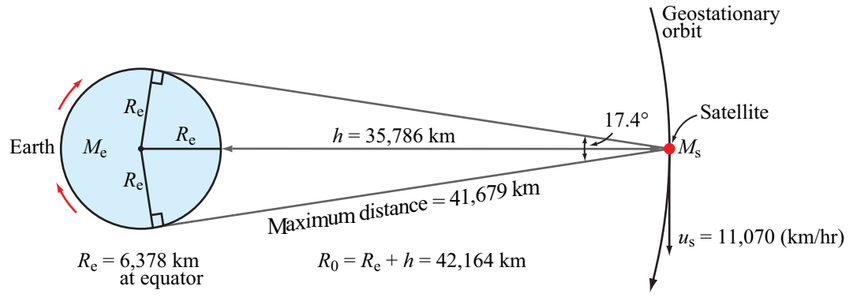



图 10-3 绕地球轨道运行的质量为 $m_s$ 的卫星。对于地球静止轨道，卫星与地球中心之间的距离 $R_0$ 应为 42,164 公里。在赤道，这相当于距地球表面 35,786 公里的高度。

$$F_c = \frac{M_s u_s^2}{R_0} = M_s \omega^2 R_0, \tag{10.2}$$

其中 $G = 6.67 \times 10^{-11}$ N·m$^2$/kg$^2$ 为万有引力常数，$M_e = 5.98 \times 10^{24}$ kg 为地球质量，$R_0$ 为卫星与地球中心之间的距离，$u_s$ 为卫星速度。对于旋转物体，$u_s = \omega R_0$，其中$\omega$是其角速度。为了使卫星保持在轨道上，作用在其上的两个相反的力必须大小相等，或者

$$G\frac{M_s M_e}{R_0^2} = M_s \omega^2 R_0, \tag{10.3}$$

得出 $R_0$ 的解：

$$R_0 = \left(\frac{GM_e}{\omega^2}\right)^{1/3}. \tag{10.4}$$

为了相对于地球表面保持静止，卫星的角速度必须与地球自身绕其自身轴的角速度相同。因此，

$$\omega = \frac{2\pi}{T}, \tag{10.5}$$

其中 T 是一个恒星日的周期（以秒为单位）。考虑到地球绕太阳的自转，一个恒星日等于 23 小时 56 分 4.1 秒。使用方程式（10.5）(10.4) 给出

$$R_0 = \left(\frac{GM_e T^2}{4\pi^2}\right)^{1/3}, \tag{10.6}$$

使用 T 、$M_e$ 和 G 的数值，我们得到结果 $R_0 = 42{,}164$ 公里。减去地球赤道处的平均半径 6,378 公里，得出距地球表面 786 公里的高度 $h = 35$。

从地球静止轨道来看，地球所成的角度为 17.4°，沿赤道覆盖了约 18,000 公里的弧线，对应于约 160° 的经度角。利用地球赤道上空静止轨道上的三颗等距卫星，可以实现整个赤道平面的完整全球覆盖，并且三颗卫星的波束之间存在显着重叠。关于两极的覆盖范围，全球波束可以到达赤道两侧纬度高达 81° 的地球站。

并非所有卫星通信系统都使用地球静止轨道上的航天器。事实上，由于发射机功率限制或其他考虑，有时需要在低得多的高度进行操作。在这种情况下，卫星被放置在高椭圆轨道上（以满足开普勒定律），因此轨道的一部分（近地点附近）距离地球表面仅几百公里。虽然只需要三颗地球静止卫星即可提供地球表面的近乎全球覆盖，但当卫星在高椭圆轨道上运行时，需要的数量要多得多。后者的一个很好的例子是技术简介 5 中描述的全球定位系统 (GPS)。

## 10-2 卫星转发器

通信卫星充当远距离中继器；它从地球站接收上行信号，处理信号，然后将它们下行（重新传输）到它们的

**Table 10-1 Communications satellite frequency allocations.**

| Use | Downlink Frequency (MHz) | Uplink Frequency (MHz) |
|---|---|---|
| **Fixed Service** | | |
| Commercial (C-band) | 3,700–4,200 | 5,925–6,425 |
| Military (X-band) | 7,250–7,750 | 7,900–8,400 |
| Commercial (K-band) | | |
| Domestic (USA) | 11,700–12,200 | 14,000–14,500 |
| International | 10,950–11,200 | 27,500–31,000 |
| **Mobile Service** | | |
| Maritime | 1,535–1,542.5 | 1,635–1,644 |
| Aeronautical | 1,543.5–1,558.8 | 1,645–1,660 |
| **Broadcast Service** | | |
| | 2,500–2,535 | 2,655–2,690 |
| | 11,700–12,750 | |
| **Telemetry, Tracking, and Command** | | |
| 137–138, 401–402, 1,525–1,540 | | |

intended Earth destinations. The International Telecommunication Union has allocated specific bands for satellite communications (**Table 10-1**). Of these, the bands used by the majority of U.S. commercial satellites for domestic communications are the ***4/6 GHz band*** (3.7 to 4.2 GHz downlink and 5.925 to 6.425 GHz uplink) and the ***12/14 GHz band*** (11.7 to 12.2 GHz downlink and 14.0 to 14.5 GHz uplink). Each uplink and downlink segment has been allocated 500 MHz of bandwidth. By using different frequency bands for Earth-to-satellite uplink segment and for satellite-to-Earth downlink segments, the same antennas can be used for both functions while simultaneously guarding against interference between the two signals. The downlink segment commonly uses a lower-frequency carrier than the uplink segment because lower frequencies suffer lower attenuation by Earth's atmosphere, thereby easing the requirement on satellite output power.

We shall use the 4/6 GHz band as a model to discuss the satellite-repeater operation, while keeping in mind that the functional configuration of the repeater is basically the same regardless of which specific communication band is used.

**Figure 10-4** shows a generalized block diagram of a typical 12-channel repeater. The path of each channel—from the point of reception by the antenna, transfer through the repeater, and final retransmission through the antenna—is called a ***transponder***. The available 500 MHz bandwidth is allocated to 12 channels (transponders) of 36 MHz bandwidth per channel and 4 MHz separation between channels. The basic functions of a transponder are (a) isolation of neighboring radio frequency (RF) channels, (b) frequency translation, and (c) amplification. With ***frequency-division multiple access*** (***FDMA***)—one of the schemes commonly used for information transmission—each transponder can accommodate thousands of individual telephone channels within its 36 MHz of bandwidth (telephone speech signals require a minimum bandwidth of 3 kHz, so frequency spacing is nominally 4 kHz per telephone channel), several TV channels (each requiring a bandwidth of 6 MHz), millions of bits of digital data, or combinations of all three.

When the same antenna is used for both transmission and reception, a ***duplexer*** is used to perform the signal separation. Many types of duplexers are available, but among the simplest to understand is the circulator shown in **Fig. 10-5**. A ***circulator*** is a three-port device that uses a ferrite material placed in a magnetic field induced by a permanent magnet to achieve power flow from ports 1 to 2, 2 to 3, and 3 to 1, but not in the reverse directions. With the antenna connected to port 1, the received signal is channeled only to port 2; if port 2 is properly matched to the band-pass filter, no part of the received signal is reflected from port 2 to 3. Similarly, the transmitted signal connected to port 3 is channeled by the circulator to port 1 for transmission by the antenna.

Following the duplexer shown in **Fig. 10-4**, the received signal passes through a receiver bandpass filter that ensures isolation of the received signal from the transmitted signal. The receiver filter covers the bandwidth from 5.925 to 6.425 GHz, which encompasses the cumulative bandwidths of all 12 channels; the first received channel extends from 5,927 to 5,963 MHz, the second one from 5,967 to 6,003 MHz, and so on until the twelfth channel, which covers the range from 6,367 to 6,403 MHz. Tracing the signal path, the next subsystem is the wideband receiver, which consists of three elements: a low-noise wideband amplifier, a frequency translator, and an output amplifier. The frequency translator consists of a stable local oscillator, which generates a signal at frequency $f_0 = 10{,}105$ MHz, connected to a nonlinear microwave mixer. The mixer serves to convert the frequency $f_r$ of the received signal (which covers the range from 5,927 to 6,403 MHz) to a lower-frequency signal $f_t = f_0 - f_r$. Thus, the lower end of the received signal frequency band gets converted from 5,927 to 4,178 MHz and the upper end gets converted from 6,403 to 3,702 MHz. This translation results in 12 channels with new frequency ranges, but these signals carry the same information (modulation) that was present in the received signals. In principle, the receiver output signal can now be further amplified and then channeled to the antenna through the duplexer for transmission back to Earth. Instead, the receiver output signal is separated into the 12 transponder channels through a ***multiplexer*** followed by a bank of narrow band-pass





表10-1 通信卫星频率分配。

| Use | Downlink Frequency (MHz) | Uplink Frequency (MHz) |
|---|---|---|
| **Fixed Service** | | |
| Commercial (C-band) | 3,700–4,200 | 5,925–6,425 |
| Military (X-band) | 7,250–7,750 | 7,900–8,400 |
| Commercial (K-band) | | |
| Domestic (USA) | 11,700–12,200 | 14,000–14,500 |
| International | 10,950–11,200 | 27,500–31,000 |
| **Mobile Service** | | |
| Maritime | 1,535–1,542.5 | 1,635–1,644 |
| Aeronautical | 1,543.5–1,558.8 | 1,645–1,660 |
| **Broadcast Service** | | |
| | 2,500–2,535 | 2,655–2,690 |
| | 11,700–12,750 | |
| **Telemetry, Tracking, and Command** | | |
| 137–138, 401–402, 1,525–1,540 | | |

预定的地球目的地。国际电信联盟为卫星通信分配了特定频段（表10-1）。其中，大多数美国商业卫星用于国内通信的频段是4/6 GHz频段（3.7至4.2 GHz下行链路和5.925至6.425 GHz上行链路）和12/14 GHz频段（11.7至12.2 GHz下行链路和14.0至14.5 GHz上行链路）。每个上行链路和下行链路段被分配了500 MHz的带宽。通过对地球到卫星上行链路段和卫星到地球下行链路段使用不同的频段，相同的天线可以用于这两种功能，同时防止两个信号之间的干扰。下行链路段通常使用比上行链路段更低频率的载波，因为较低频率受到地球大气层的衰减较小，从而减轻了对卫星输出功率的要求。

我们将使用 4/6 GHz 频段作为模型来讨论卫星中继器的操作，同时记住，无论使用哪个特定通信频段，中继器的功能配置基本相同。

图 10-4 显示了典型 12 通道中继器的通用框图。每个通道的路径（从天线接收点，通过中继器传输，最后通过天线重传）称为转发器。可用的 500 MHz 带宽分配给 12 个通道（转发器），每个通道 36 MHz 带宽，通道之间间隔 4 MHz。基本的

应答器的功能包括 (a) 相邻射频 (RF) 通道的隔离、(b) 频率转换和 (c) 放大。通过频分多址（FDMA）（信息传输常用的方案之一），每个转发器可以在其 36 MHz 带宽内容纳数千个单独的电话通道（电话语音信号需要的最小带宽为 3 kHz，因此每个电话通道的频率间隔名义上为 4 kHz）、多个电视通道（每个通道需要 6 MHz 的带宽）、数百万比特的数字数据或三者的组合。

当同一天线用于发送和接收时，使用双工器来执行信号分离。双工器有多种类型，但最容易理解的是图 10-5 所示的循环器。环行器是一种三端口器件，它使用放置在永磁体感应磁场中的铁氧体材料来实现从端口 1 到 2、2 到 3 和 3 到 1 的功率流，但不能反向流动。当天线连接到端口 1 时，接收到的信号仅传输到端口 2；如果端口 2 与带通滤波器正确匹配，接收信号的任何部分都不会从端口 2 反射到端口 3。同样，连接到端口 3 的发射信号由循环器引导到端口 1，以便通过天线进行发射。

在图 10-4 所示的双工器之后，接收信号通过接收器带通滤波器，确保接收信号与发射信号的隔离。接收器滤波器覆盖 5.925 至 6.425 GHz 的带宽，其中包含所有 12 个通道的累积带宽；第一个接收信道从 5,927 MHz 延伸到 5,963 MHz，第二个接收信道从 5,967 MHz 延伸到 6,003 MHz，以此类推，直到第十二个信道，覆盖从 6,367 MHz 到 6,403 MHz 的范围。沿着信号路径，下一个子系统是宽带接收器，它由三个元件组成：低噪声宽带放大器、频率转换器和输出放大器。频率转换器由一个稳定的本地振荡器组成，该振荡器生成频率为 $f_0 = 10{,}105$ MHz 的信号，并连接到非线性微波混频器。混频器用于将接收信号的频率 $f_r$（覆盖范围为 5,927 至 6,403 MHz）转换为低频信号 $f_t = f_0 - f_r$。因此，接收信号频带的下端从 5,927 MHz 转换为 4,178 MHz，上端从 6,403 MHz 转换为 3,702 MHz。这种转换会产生具有新频率范围的 12 个通道，但这些信号携带与接收信号中存在的相同信息（调制）。原则上，接收器输出信号现在可以进一步放大，然后通过双工器传输到天线，然后传输回地球。相反，接收器输出信号通过复用器分为 12 个应答器通道，然后是一组窄带通

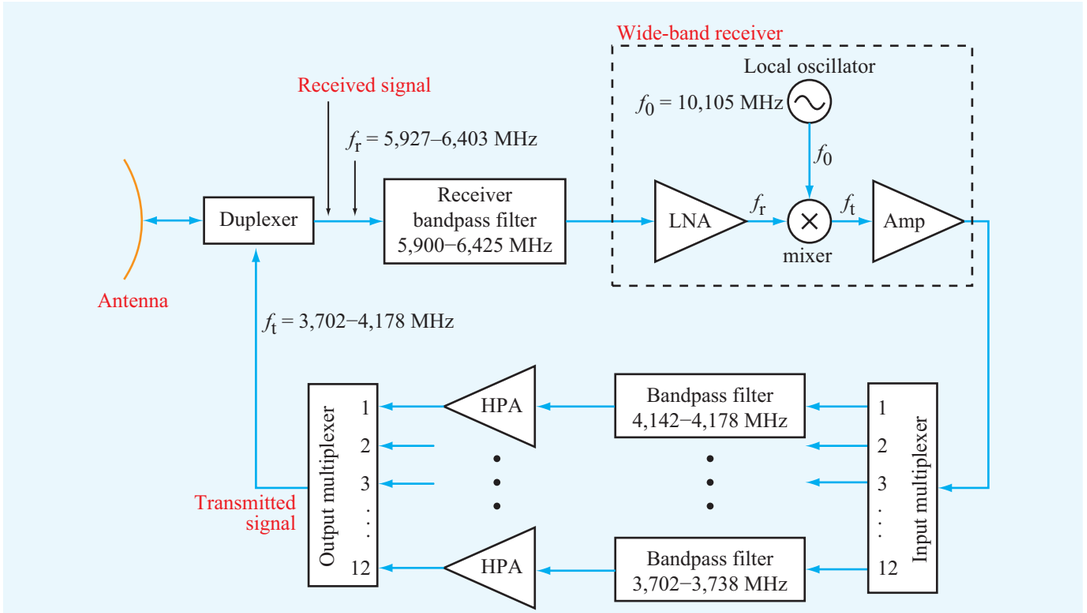



**Figure 10-4** Elements of a 12-channel (transponder) communications system.

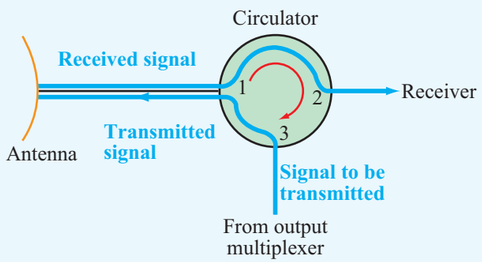



**Figure 10-5** Basic operation of a ferrite circulator.

filters with each covering the bandwidth of one transponder channel. Each of the 12 channels is amplified by its own ***high-power amplifier*** (***HPA***), and then the 12 channels are combined by another multiplexer that feeds the combined spectrum into the duplexer. This channel separation and recombination process is used as a safety measure against losing all 12 channels should a high-power amplifier experience total failure or degradation in performance.

The information carrying capacity of a satellite repeater can be doubled from 12 to 24 channels over the same 500 MHz bandwidth by using ***polarization diversity***. Instead of transmitting one channel of information over channel 1 (5,927 to 5,963 MHz), for example, the ground station transmits to the satellite two signals carrying different information and covering the same frequency band, but with different antenna polarization configurations, such as right-hand circular (RHC) and left-hand circular (LHC) polarizations. The satellite antenna is equipped with a feed arrangement that can receive each of the two circular polarization signals individually with negligible interference between them. Two duplexers are used in this case: one connected to the RHC polarization feed and another connected to the LHC polarization feed, as illustrated in **Fig. 10-6**.





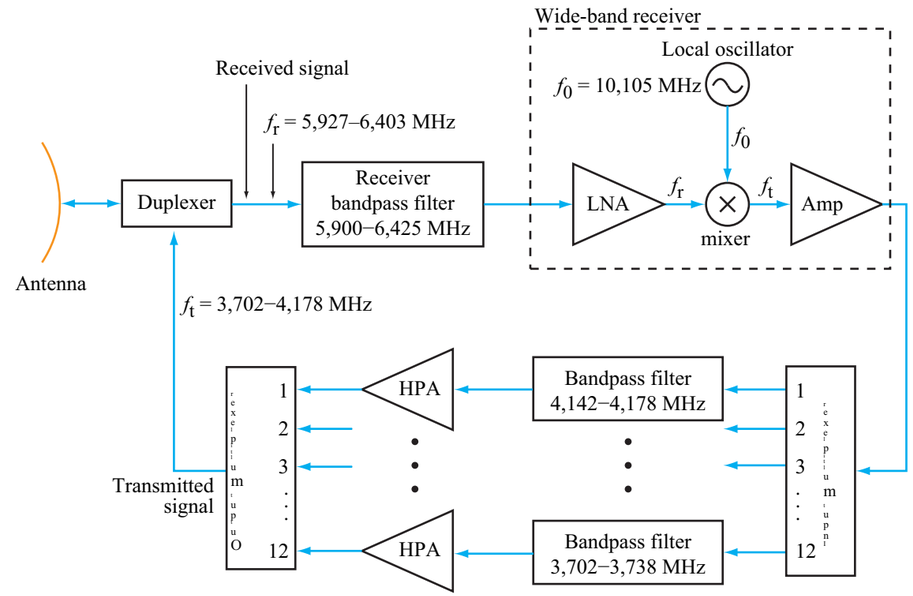



图 10-4 12 通道（应答器）通信系统的元件。

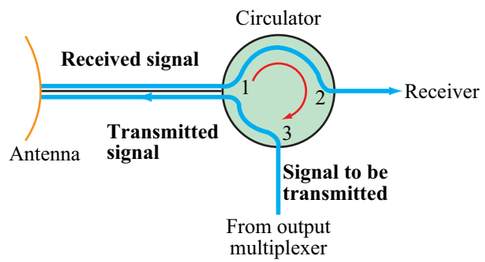



图 10-5 铁氧体环行器的基本操作。

每个滤波器都覆盖一个转发器通道的带宽。 12 个通道中的每一个通道均由其自己的高功率放大器 (HPA) 进行放大，然后将 12 个通道组合起来

由另一个多路复用器将组合频谱馈入双工器。这种通道分离和重组过程被用作一种安全措施，防止在高功率放大器出现完全故障或性能下降时丢失所有 12 个通道。

通过使用极化分集，卫星中继器的信息承载能力可以在相同的 500 MHz 带宽上从 12 个通道增加一倍到 24 个通道。例如，地面站不是通过信道 1（5,927 至 5,963 MHz）传输一个信道的信息，而是向卫星发送两个携带不同信息并覆盖相同频段的信号，但具有不同的天线极化配置，例如右旋圆（RHC）和左旋圆（LHC）极化。卫星天线配备了馈电装置，可以单独接收两个圆极化信号中的每一个，并且它们之间的干扰可以忽略不计。在这种情况下使用两个双工器：一个连接到 RHC 偏振馈电，另一个连接到 LHC 偏振馈电，如图 10-6 所示。

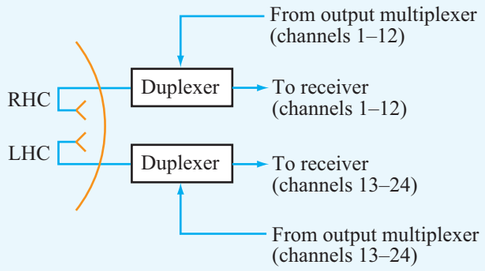



**Figure 10-6** Polarization diversity is used to increase the number of channels from 12 to 24.

## 10-3 Communication-Link Power Budget

The uplink and downlink segments of a satellite communication link (**Fig. 10-7**) are each governed by the ***Friis transmission formula*** (Section 9-6), which states that the power $P_r$ received by an antenna with gain $G_r$ due to the transmission of power $P_t$ by an antenna with gain $G_t$ at a range $R$ is given by

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2. \tag{10.7}$$

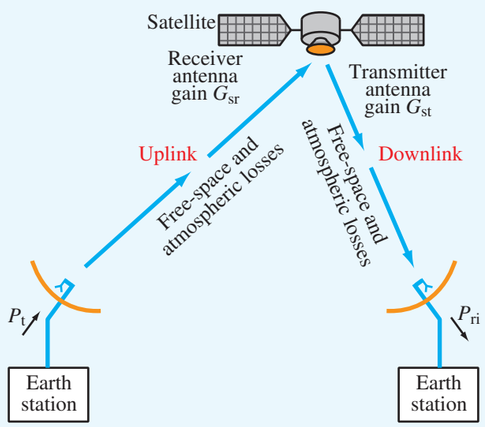



**Figure 10-7** Satellite transponder.

This expression applies to a lossless medium, such as free space. To account for attenuation by clouds and rain in Earth's atmosphere (when present along the propagation path), as well as absorption by certain atmospheric gases (primarily oxygen and water vapor), we rewrite Eq. (10.7) as

$$P_{ri} = \Upsilon(\theta)\, P_r = \Upsilon(\theta)\, P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2. \tag{10.8}$$

Now, $P_{ri}$ represents the input power at the receiver with atmospheric losses taken into account, and $\Upsilon(\theta)$ is the one-way ***transmissivity*** of the atmosphere at zenith angle $\theta$. In addition to its dependence on $\theta$, $\Upsilon(\theta)$ is a function of the frequency of the communication link and the rain-rate conditions along the propagation path. At frequencies below 10 GHz, which include the 4/6 GHz band allocated for satellite communication, absorption by atmospheric gases is very small, as is attenuation due to clouds and rain. Consequently, the magnitude of $\Upsilon(\theta)$ is typically on the order of 0.5 to 1 for most conditions. A transmissivity of 0.5 means that twice as much power needs to be transmitted (compared to the free-space case) in order to receive a specified power level. Among the various sources of atmospheric attenuation, the most serious is rainfall, and its attenuation coefficient increases rapidly with increasing frequency. Consequently, atmospheric attenuation assumes greater importance with regard to transmitter power requirements as the communication-system frequency is increased toward higher bands in the microwave region.

The noise appearing at the receiver output, $P_{no}$, consists of three contributions: (1) noise internally generated by the receiver electronics, (2) noise picked up by the antenna due to external sources, including emission by the atmosphere, and (3) noise due to thermal emission by the antenna material. The combination of all noise sources can be represented by an equivalent ***system noise temperature***, $T_{sys}$, defined such that

$$P_{no} = G_{rec} K T_{sys} B, \tag{10.9}$$

where $K$ is Boltzmann's constant, and $G_{rec}$ and $B$ are the receiver power gain and bandwidth, respectively. This output noise level is the same as would appear at the output of a noise-free receiver with input noise level:

$$P_{ni} = \frac{P_{no}}{G_{rec}} = K T_{sys} B. \tag{10.10}$$

The ***signal-to-noise ratio*** is defined as the ratio of the signal power to the noise power *at the input of an equivalent noise-free receiver*. Hence,

$$S_n = \frac{P_{ri}}{P_{ni}} = \frac{\Upsilon(\theta)\, P_t G_t G_r}{K T_{sys} B} \left(\frac{\lambda}{4\pi R}\right)^2. \tag{10.11}$$





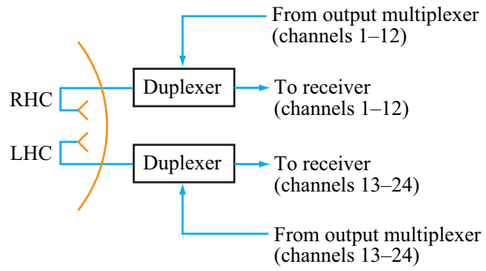



图10-6 极化分集用于将通道数从12个增加到24个。

## 10-3 通信链路功率预算

卫星通信链路的上行链路和下行链路段（图 10-7）均由 Friis 传输公式（第 9-6 节）控制，该公式指出，由于增益为 $G_t$ 的天线在范围 R 上发射功率 $P_t$，增益为 $G_r$ 的天线接收到的功率 $P_r$ 由下式给出：

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2. \tag{10.7}$$

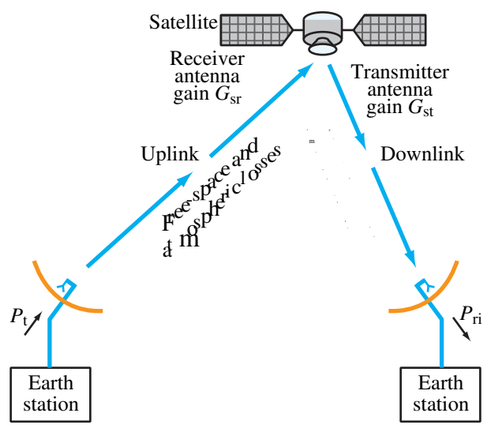



图 10-7 卫星转发器。

此表达式适用于无损介质，例如自由空间。为了考虑地球大气中云和雨的衰减（当沿着传播路径存在时），以及某些大气气体（主要是氧气和水蒸气）的吸收，我们重写了方程：(10.7) 作为

$$P_{ri} = \Upsilon(\theta)\, P_r = \Upsilon(\theta)\, P_t G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2. \tag{10.8}$$

现在，$P_{ri}$ 表示考虑了大气损耗的接收器处的输入功率，而 $\Upsilon(\theta)$ 是天顶角 θ 处大气的单向透射率。除了依赖于 θ 之外，$\Upsilon(\theta)$ 也是通信链路频率和传播路径上降雨率条件的函数。在 10 GHz 以下的频率（包括分配给卫星通信的 4/6 GHz 频段），大气气体的吸收非常小，云和雨造成的衰减也非常小。因此，在大多数情况下，$\Upsilon(\theta)$ 的大小通常约为 0.5 到 1。 0.5 的传输率意味着需要传输两倍的功率（与自由空间情况相比）才能接收指定的功率电平。在大气衰减的各种来源中，最严重的是降雨，其衰减系数随着频率的增加而迅速增加。因此，随着通信系统频率向微波区域的更高频段增加，大气衰减对于发射机功率要求变得更加重要。

接收器输出处出现的噪声 $P_{no}$ 由三个部分组成：(1) 接收器电子设备内部产生的噪声；(2) 由于外部源（包括大气发射）而由天线拾取的噪声；(3) 由于天线材料的热发射而产生的噪声。所有噪声源的组合可以用等效系统噪声温度 $T_{sys}$ 表示，定义为

$$P_{no} = G_{rec} K T_{sys} B, \tag{10.9}$$

其中 K 是玻尔兹曼常数，$G_{rec}$ 和 B 分别是接收器功率增益和带宽。该输出噪声电平与具有输入噪声电平的无噪声接收器的输出端出现的噪声电平相同：

$$P_{ni} = \frac{P_{no}}{G_{rec}} = K T_{sys} B. \tag{10.10}$$

信噪比定义为等效无噪声接收器输入端的信号功率与噪声功率之比。因此，

$$S_n = \frac{P_{ri}}{P_{ni}} = \frac{\Upsilon(\theta)\, P_t G_t G_r}{K T_{sys} B} \left(\frac{\lambda}{4\pi R}\right)^2. \tag{10.11}$$

The performance of a communication system is governed by two sets of issues. The first encompasses the signal-processing techniques used to encode, modulate, combine, and transmit the signal at the transmitter end and to receive, separate, demodulate, and decode the signal at the receiver end. The second set encompasses the gains and losses in the communication link, and they are represented by the signal-to-noise ratio $S_n$. For a given set of signal-processing techniques, $S_n$ determines the quality of the received signal, such as the bit error rate in digital data transmission and sound and picture quality in audio and video transmissions. Very high-quality signal transmission requires very high values of $S_n$; in broadcast-quality television by satellite, some systems are designed to provide values of $S_n$ exceeding 50 dB (or a factor of $10^5$).

The performance of a satellite link depends on the composite performance of the uplink and downlink segments. If either segment performs poorly, the composite performance will be poor, regardless of how good the performance of the other segment is.

## 10-4 Antenna Beams

Whereas most Earth-station antennas are designed to provide highly directive beams (to avoid interference effects), the satellite antenna system is designed to produce beams tailored to match the areas served by the satellite. For global coverage, beamwidths of 17.4° are required. In contrast, for transmission to and reception from a small area, beamwidths on the order of 1° or less may be needed (**Fig. 10-8**).

An antenna with a beamwidth $\beta$ of 1° would produce a spot beam on Earth covering an area approximately 630 km in diameter.

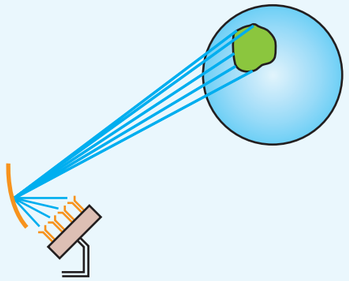
Beam size has a direct connection to antenna gain and, in turn, to transmitter power requirements. Antenna gain $G$ is related to the directivity $D$ by $G = \xi D$, where $\xi$ is the radiation efficiency and $D$ is related to the beamwidth $\beta$ by the approximate expression given by Eq. (9.26). For a circular beam,

$$G = \xi \frac{4\pi}{\beta^2}, \tag{10.12}$$

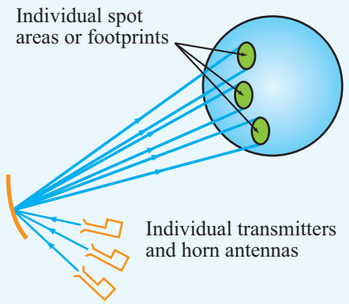
where $\beta$ is in radians. For a lossless antenna ($\xi = 1$), a global beam with $\beta = 17.4°$ (= 0.3 rad) corresponds to a gain $G = 136$ or 21.3 dB. A narrow 1° beam, on the other hand, corresponds to an antenna gain of 41,253 or 46.2 dB.

To accommodate the various communication functions associated with satellite systems, four main types of antennas are used.*

*R. G. Meadows and A. J. Parsons, *Satellite Communications*, Hutchinson Publishers, London, 1989.



(a) Zone coverage




(b) Multispot beams

**Figure 10-8** Spot and multibeam satellite antenna systems for coverage of defined areas on Earth's surface.

1. Dipoles and helices at VHF and UHF are used for telemetry, tracking, and command functions.

2. Horns and relatively small parabolic dishes (with diameters on the order of a few centimeters) to produce wide-angle beams for global coverage.

3. Parabolic dishes fed by one or more horns provide a beam for zone coverage (**Fig. 10-8(a)**) or multiple spot beams (**Fig. 10-8(b)**).

4. Antenna arrays consisting of many individual radiating elements are used to produce multispot beams and for beam steering and scanning.

440 CHAPTER 10 SATELLITE COMMUNICATION SYSTEMS AND RADAR SENSORS

通信系统的性能由两组问题决定。第一类包括信号处理技术，用于在发送端对信号进行编码、调制、组合和发送，以及在接收端接收、分离、解调和解码信号。第二组包含通信链路中的增益和损耗，它们由信噪比 $S_n$ 表示。对于一组给定的信号处理技术，$S_n$ 决定接收信号的质量，例如数字数据传输中的误码率以及音频和视频传输中的声音和图像质量。非常高质量的信号传输需要非常高的$S_n$值；在卫星广播质量电视中，一些系统设计为提供超过 50 dB（或 $10^5$ 因子）的 $S_n$ 值。

卫星链路的性能取决于上行链路和下行链路段的综合性能。如果任一部分表现不佳，则无论其他部分的表现有多好，综合表现也将很差。

## 10-4 天线波束

大多数地球站天线的设计目的是提供高度定向的波束（以避免干扰效应），而卫星天线系统的设计目的是产生适合卫星服务区域的波束。对于全球覆盖，需要 17.4° 的波束宽度。相反，对于小区域的传输和接收，可能需要 1° 或更小的波束宽度（图 10-8）。

波束宽度 β 为 1° 的天线将在地球上产生覆盖直径约 630 公里区域的点波束。

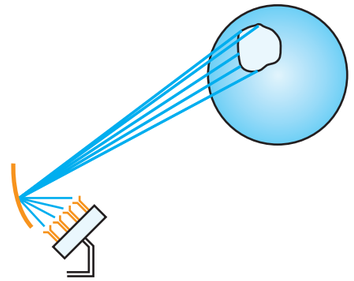
波束尺寸与天线增益直接相关，进而与发射机功率要求直接相关。天线增益 G 通过 $G = \xi D$ 与方向性 D 相关，其中 ξ 是辐射效率，D 通过等式给出的近似表达式与波束宽度 β 相关。（9.26）。对于圆形波束，

$$G = \xi \frac{4\pi}{\beta^2}, \tag{10.12}$$

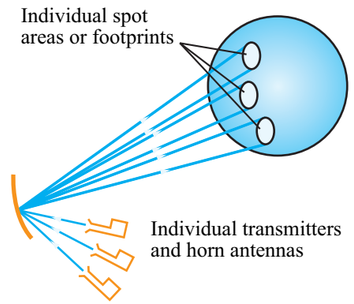
其中 β 的单位是弧度。对于无损天线 ($\xi = 1$)，$\beta = 17.4°$ (= 0.3 rad) 的全局波束对应于增益 G = 136 或 21.3 dB。另一方面，窄 1° 波束对应于 41,253 或 46.2 dB 的天线增益。

为了适应与卫星系统相关的各种通信功能，使用了四种主要类型的天线。*

*R. G. Meadows and A. J. Parsons, *Satellite Communications*, Hutchinson Publishers, London, 1989.



(a) Zone coverage




(b) Multispot beams

图 10-8 用于覆盖地球表面指定区域的点和多波束卫星天线系统。

1. VHF 和 UHF 的偶极子和螺旋用于遥测、跟踪和命令功能。 2. 喇叭和相对较小的抛物面天线（直径约为几厘米）可产生覆盖全球的广角波束。 3. 由一个或多个喇叭馈送的抛物面天线提供区域覆盖波束（图 10-8(a)）或多点波束（图 10-8(b)）。 4. 由许多单独的辐射元件组成的天线阵列用于产生多点波束并用于波束控制和扫描。